

SPI Slave IP for IHP SG13G2 — Verification & Signoff Report



Mode-0 and all-modes (CPOL/CPHA) variants · 1.2 V
(sg13g2_stdcell) and 3.3 V (sg13g2_stdcell_hv)
implementations · RTL, host-level, RTL-to-GDS, DRC and
LVS

Koen Van Caekenberghe, Ph.D. — ChipDesign B.V. —
info@chipdesign.be

2026-08-16 (rev. 2: thick-oxide implementations added)

1 Scope and summary

This report documents the verification and physical signoff of the `spi_slave` IP block for the IHP SG13G2 0.13 μm BiCMOS open PDK, in its two variants:

- **Mode-0 variant** (repository root) — SPI mode 0 only (CPOL=0, CPHA=0).
- **All-modes variant** (allmodes/) — SPI modes 0–3, selected by two configuration pins CPOL/CPHA; designed to interface both a Raspberry Pi (spidev master) and an Infineon AURIX microcontroller (QSPI master).

Each variant is implemented **twice**: on the PDK's 1.2 V thin-oxide library `sg13g2_stdcell` (sections 5–7) and on the 3.3 V thick-oxide library `sg13g2_stdcell_hv` (section 8), for four GDS deliverables in total. All verification steps pass for both variants and both libraries:

Verification step	Mode-0 variant	All-modes variant
RTL functional testbench	pass	pass (all 4 modes)
B2B RTL vs. synthesised netlist	pass, bit-identical	pass, bit-identical (all 4 modes)
Raspberry Pi master model (RTL + netlist)	n/a	pass (all 4 modes)
AURIX QSPI master model, full register sweep (RTL + netlist)	n/a	pass (all 4 modes)
RTL-to-GDS flow, 1.2 V (LibreLane v3.1.0 Classic, 70 steps)	complete	complete
Static timing, 1.2 V (3 corners, setup + hold)	0 violations	0 violations
RTL-to-GDS flow, 3.3 V (sg13g2_stdcell_hv)	complete	complete

Verification step	Mode-0 variant	All-modes variant
Static timing, 3.3 V (typical corner, setup + hold)	0 violations	0 violations
Gate-level simulation, 3.3 V netlist	pass	pass (all 4 modes)
Antenna check (both libraries)	0 violations	0 violations
LVS (both libraries)	match uniquely	match uniquely
Signoff DRC, IHP KLayout full deck (all four core macros)	clean ¹	clean ¹

¹ Apart from eight single-marker chip-level metal-density and filler-presence checks that fire on any unfilled macro (section 6.3); these are resolved by foundry density fill at tapeout.

The signoff investigation additionally uncovered a **defect in the LibreLane v3 KLayout.SealRing step** (section 7): the inserted seal ring physically overlaps the routed core and carries an incorrect EdgeSeal marker. The repository GDS deliverables are therefore the pre-seal-ring stream-outs; a seal ring must be added correctly at chip assembly.

2 Design description

2.1 Architecture (both variants)

The `spi_slave` module exposes eight 8-bit registers over SPI and provides a secondary parallel read port (`Add2_in/Data2_out`) for concurrent on-chip register readback independent of the SPI bus.

Feature	Value
Register map	8 × 8-bit, addresses 0x00–0x07
Command byte	bit 7 = R/nW (1 = read, 0 = write); bits 6:0 = address
Out-of-range read	returns 0xFF
Reset	active-low asynchronous <code>iRST_N</code> ; registers reset to 0x0A...0x30
Debug port	<code>debug[7:0]</code> — sticky OR of FSM states visited since reset
FSM	5-state Moore: IDLE → COMMAND → READ/ WRITE → END

All asynchronous SPI inputs pass through a two-flip-flop metastability synchroniser before entering the system-clock (`clk`) domain; SPI clock edges are detected by comparing two successive synchronised samples.

2.2 All-modes extension

The all-modes variant adds two configuration inputs and generalises the edge handling. The synchronised SCK is normalised with the clock polarity,

$$SCK_{norm} = SCK \oplus CPOL$$

so the FSM always sees mode-0 polarity: the leading edge is a rising edge of SCK_{norm} , the trailing edge a falling edge. The clock phase then selects the roles of the two edges:

Mode	CPOL	CPHA	SCK idle	Sample edge (both sides)	Slave drives MISO on
0	0	0	low	leading (rising)	trailing (falling)
1	0	1	low	trailing (falling)	leading (rising)
2	1	0	high	leading (falling)	trailing (rising)
3	1	1	high	trailing (rising)	leading (falling)

CPOL/CPHA pass through the same two-flop synchroniser and are latched only while the FSM is in IDLE, so a transaction always runs with a consistent mode pair; the pins may be strap resistors or host GPIOs, stable before SSEL assertion.

Because SCK is oversampled by Clk , the synchroniser and edge detector add up to three Clk cycles of latency between a physical SCK edge and the MISO update. With the constraint $f_{SCK} \leq f_{Clk}/8$ the slave output is guaranteed stable at the master's sampling edge with margin; at the implemented $f_{Clk} = 100$ MHz this permits $f_{SCK} \leq 12.5$ MHz.

3 RTL functional verification

3.1 Mode-0 variant

Testbench `tb_spi_slave.v` (Icarus Verilog ≥ 11) executes three transactions: Tx1 reads Register 1 (expects reset value 0x0B), Tx2 writes 0xFF to Register 3, Tx3 reads Register 3 back. Result:

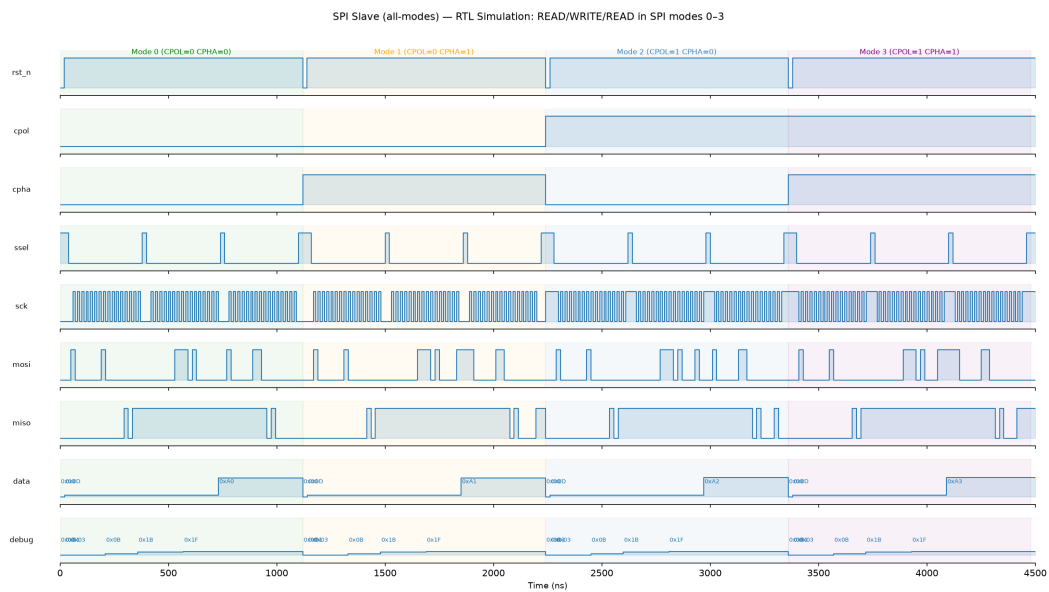
```
PASS 1100: miso_rx1=0x0B data=0xFF miso_rx3=0xFF debug=0x1F
```

`debug = 0x1F` confirms all five FSM states were visited; `data = 0xFF` confirms the write is also visible on the parallel read port.

3.2 All-modes variant

Testbench `allmodes/tb_spi_slave.v` repeats a read/write/read-back sequence in each of the four SPI modes, with a DUT reset between modes and a mode-correct master model (data set up before the leading edge for CPHA=0, changed on the leading edge for CPHA=1):

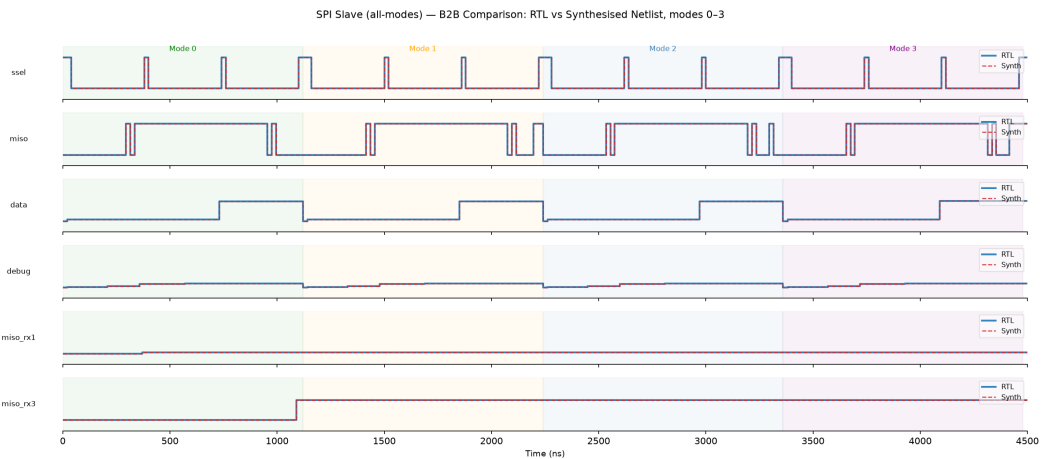
```
MODE 0 (CPOL=0 CPHA=0) 1120: miso_rx1=0x0b data=0xa0 miso_rx3=0xa0 debug=0x1f
MODE 1 (CPOL=0 CPHA=1) 2240: miso_rx1=0x0b data=0xa1 miso_rx3=0xa1 debug=0x1f
MODE 2 (CPOL=1 CPHA=0) 3360: miso_rx1=0x0b data=0xa2 miso_rx3=0xa2 debug=0x1f
MODE 3 (CPOL=1 CPHA=1) 4480: miso_rx1=0x0b data=0xa3 miso_rx3=0xa3 debug=0x1f
PASS: all checks passed in all 4 SPI modes
```



All-modes RTL simulation — read/write/read-back in SPI modes 0–3, with mode windows shaded

3.3 Back-to-back RTL vs. synthesised netlist

Each variant's testbench is re-run against its Yosys-synthesised structural netlist (`flow/spi_slave_synth.v`, generic cells). For both variants every checked signal is bit-for-bit identical between RTL and netlist across the entire simulation, and the same PASS line is printed.



B2B overlay for the all-modes variant: RTL (blue) vs. synthesised netlist (red dashed) across all four modes — traces overlap completely

4 Host-level verification (all-modes variant)

Two additional testbenches model the intended host masters. Both run in all four SPI modes against the RTL **and** the synthesised netlist (eight runs in total, all passing), with the DUT clocked at its implemented 100 MHz.

4.1 Raspberry Pi master model

`tb_spi_slave_rpi.v` models the BCM283x/BCM2711 SPI controller as driven by `spidev's xfer2()`: CE0 asserted for the whole 2-byte buffer with about one SCK period of setup, bytes clocked back-to-back with no inter-byte gap, MSB first, 10 MHz SCK. Per mode it checks: register read (0x0B), write + read-back (0x5A), the parallel port, an out-of-range address read (0xFF) and `debug = 0x1F`.

PASS: Raspberry Pi master model – all checks passed in all 4 SPI modes

4.2 AURIX QSPI master model

`tb_spi_slave_aurix.v` models an AURIX (TC2xx/TC3xx) QSPI channel: SLSO lead and trail delays and an inter-word idle of one SCLK period each (SLSO held asserted across the frame), 12.5 MHz SCLK — deliberately at the $f_{CLK}/8$ limit to demonstrate margin. Per mode it performs a full register-map sweep: writes a distinct value to all eight registers, reads all eight back over SPI, reads all eight through the parallel port, then checks the out-of-range read and the debug port.

PASS: AURIX QSPI master model – all checks passed in all 4 SPI modes

4.3 Host wiring summary

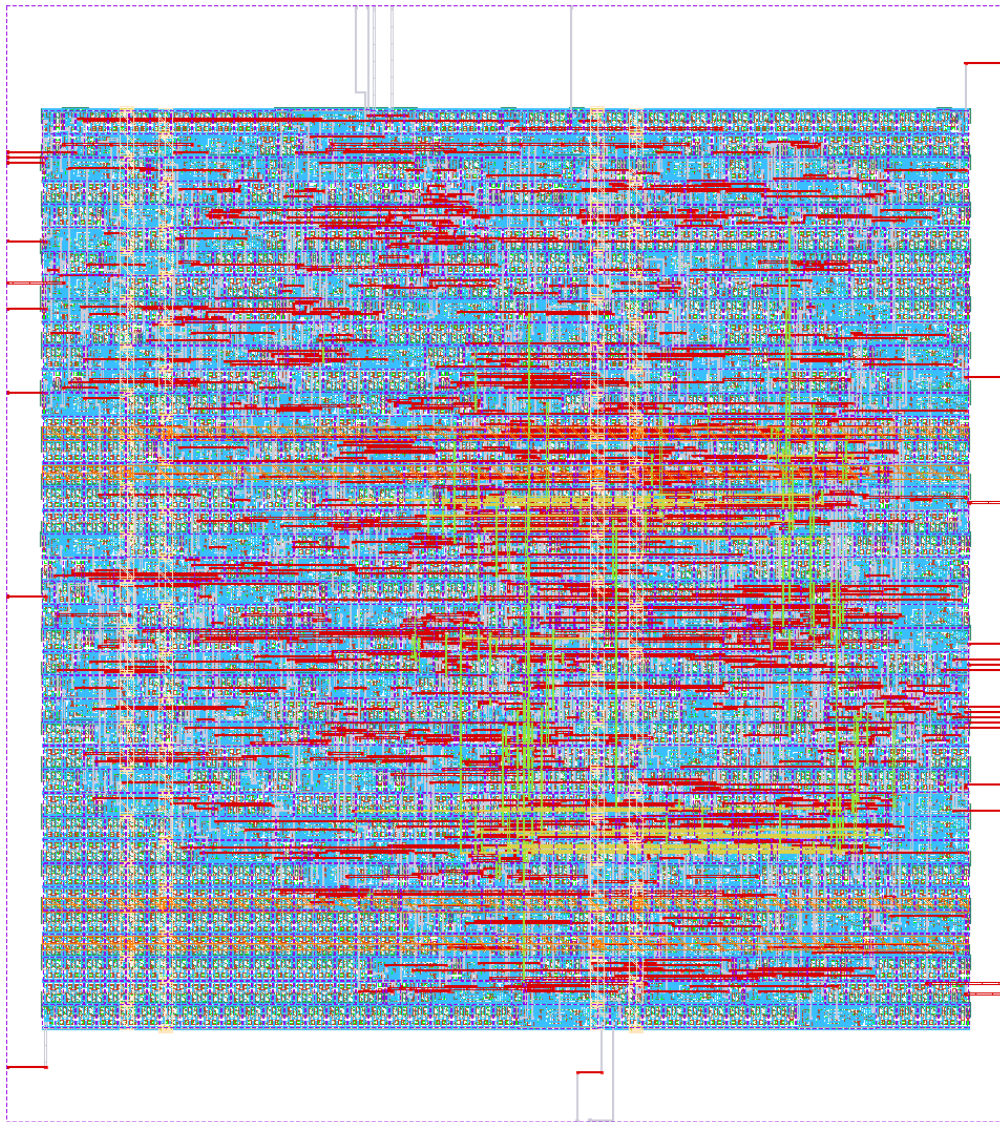
Host	Connections	Mode selection
Raspberry Pi (3.3 V, direct)	pin 19 MOSI, pin 21 MISO, pin 23 SCLK, pin 24 CE0 → SSEL	<code>spi.mode = 0...3</code> in <code>spidev</code> ; strap or GPIO on CP0L/CPHA to match
AURIX (3.3 V pad supply)	QSPI MTSR → MOSI, MRST ← MISO, SCLK → SCK, SLS0[n] → SSEL	iLLD <code>clockPolarity / shiftClock channel config</code> ; strap or GPIO on CP0L/CPHA

Both hosts must keep the chip select asserted across the command + data byte pair (`spidev xfer2` and a 2-byte QSPI exchange buffer both do), and keep $f_{SCK} \leq f_{CLK}/8$.

5 RTL-to-GDS implementation (1.2 V, `sg13g2_stdcell`)

Both variants were taken through the LibreLane v3.1.0 Classic flow (70 steps) on the IHP SG13G2 open PDK's thin-oxide standard cell library, with a 10 ns system clock constraint. Timing is analysed at three corners: slow 1.08 V 125 °C, typical 1.20 V 25 °C, fast 1.32 V –40 °C.

Metric	Mode-0 variant	All-modes variant
Flow run	RUN_2026-08-08_22-27-58	RUN_2026-08-08_21-52-37
Die size	157.78 μm × 176.50 μm	160.61 μm × 179.33 μm
Std-cell utilisation	≈ 54 %	53.5 %
Setup WNS / TNS (all corners)	0 / 0	0 / 0
Hold WNS / TNS (all corners)	0 / 0	0 / 0
Detailed-routing DRC errors	0	0
Antenna violations	0	0
Max slew / max cap violations	0 / 0	0 / 0
Max fanout violations	—	1 (buffer drives 9 loads, limit 8; no timing impact)
Routed wirelength	—	21.2 mm
Total power (tt, 1.2 V, 25 °C)	—	≈ 1.08 mW



All-modes variant — final routed layout (LibreLane render)

6 Signoff LVS and DRC

6.1 LVS

LVS is performed by the flow's signoff stage: Magic extracts a SPICE netlist from the final layout, and Netgen compares it against the post-PnR powered Verilog netlist mapped to the PDK SPICE libraries.

Design	Devices (layout = netlist)	Nets (layout = netlist)	Verdict
Mode-0 variant	597 = 597	608 = 608	Circuits match uniquely

Design	Devices (layout = netlist)	Nets (layout = netlist)	Verdict
All-modes variant	617 = 617	630 = 630	Circuits match uniquely

Zero unmatched devices, nets or pins, and zero property failures in both runs.

6.2 DRC methodology

Signoff DRC uses the IHP SG13G2 KLayout rule deck (\$PDK_ROOT/ihp-sg13g2/libs.tech/klayout/tech/drc/run_drc.py), full rule set, deep mode, identical invocation for every run — violation counts from different invocations of this deck are not comparable, so a fixed harness is used throughout. The verdict is taken from the produced .lyrdb marker database, not from the runner's exit code.

Two control runs anchor the interpretation:

1. **Pristine PDK standard cell** (sg13g2_xnor2_1 from the PDK library) through the identical harness → 7 markers, all single-count chip-level density/filler checks. Confirms the harness does not flag IHP's own cells.
2. **Pre-seal-ring stream-outs** of both designs (below).

6.3 DRC results

Layout under test	Total markers	Content
Pristine PDK std cell (control)	7	chip-level density/filler only
Mode-0 core macro (pre-seal-ring)	8	chip-level density/filler only
All-modes core macro (pre-seal-ring)	8	chip-level density/filler only
Mode-0 final GDS incl. seal ring	21,208	seal-ring defect (section 7)
All-modes final GDS incl. seal ring	15,177	seal-ring defect (section 7)

The eight core-macro markers are M1.j–M5.j, TM1.c, TM2.c (metal density windows) and AFil.g/GFil.g (active/gate filler presence) — one marker each. These are chip-level manufacturing-readiness checks that fire on any macro without dummy fill; they are resolved by foundry density fill at tapeout and are not layout defects. **Both core macros are therefore DRC-clean**, and both are shipped as the repository's spi_slave.gds deliverables.

7 LibreLane v3 seal-ring defect

The seal-ringed final GDS produced by the LibreLane v3 KLayout .SealRing step fails signoff DRC massively in both designs, with identical rule signatures. The failure was isolated experimentally:

- **The ring physically overlaps the routed core.** The step generates the ring PCell sized die – 64.4 μm and inset 32.2 μm from the die origin — inside the core area instead of surrounding it. Measured overlap between ring and core geometry (Activ + pSD + Metal1 + Cont): $\approx 2,316 \mu\text{m}^2$ in the mode-0 design's previously shipped GDS, $\approx 1,730 \mu\text{m}^2$ in the all-modes design. Ring metal crossing core routing constitutes physical shorts to the grounded seal structure; these layouts are not manufacturable.
- **The EdgeSeal (39/4) marker is a full-die rectangle** instead of the ring annulus. The DRC deck derives the seal-ring region from this layer, so the entire chip is classified as seal-ring structure and every core shape violates the Seal.* rule family — the bulk of the 15k–21k markers. Rewriting EdgeSeal as the true annulus removes this cascade but cannot repair the physical core overlap.

Because the defect is in the flow step (present identically in both designs, including the previously published mode-0 GDS), the repository deliverables were regenerated from the DRC/LVS-clean pre-seal-ring KLayout stream-outs. **The LibreLane v3 seal-ringed GDS must not be used for fabrication**; the seal ring is to be added correctly at chip assembly. Flow-level notes and the resume procedure are documented in `flow/README.md`.

8 Thick-oxide (3.3 V) implementation

Both variants were re-implemented on `sg13g2_stdcell_hv`, a thick-oxide 3.3 V rebuild of the IHP standard cells (a sibling repository with its own generation and sign-off report), through the same LibreLane v3.1.0 Classic flow at the same 10 ns clock. The flows live in `flow_hv/` and `allmodes/flow_hv/`; the signoff GDS/DEF and their DRC runs in `signoff_hv/` and `allmodes/signoff_hv/`.

8.1 Flow adaptations

Every deviation from the thin-oxide flow is a consequence of the library, documented at the point of use in `flow_hv/config.yaml`:

- **Flip-flop mapping** (`hv_dff_map.v`): `sg13g2_hv_sdfbbp_1` is the only flip-flop with layout; the four flavours the RTL produces map onto it with the scan mux reused as the clock-enable mux. Gate-level simulation (`verify_gl.sh`) is what proves this mapping right.
- **Excluded cells**: 12 characterised-but-unlaid-out flip-flops/latches, plus `sg13g2_hv_nand4_1` (its B pin has no routing track through it).

- **Metal1 routing** (RT_MIN_LAYER: Metal1): 25 library pins sit off the 0.48 μm vertical track grid, and the router needs Metal1 jogs to reach them.
- **No preventive antenna diodes**: the heuristic pass placed 1 088 diodes the design does not need (~20 % of the std-cell area); targeted repair on the nets that need it keeps the antenna check at zero.
- **Rails-only fillers**: fillers are inserted after routing, and the decap cells' Metal1 straps are invisible to the router — with Metal1 routing they can land inside the 0.18 μm Metal1 spacing of a routed wire. The fill_* cells carry Metal1 only on the rails.
- **One timing corner** (typical, 3.30 V, 25 °C): the library is characterised nowhere else, and it carries no internal-power tables, so power is not reported.

8.2 Results

Metric	Mode-0 variant	All-modes variant
Flow run	RUN_2026-08-16_13-42-07	RUN_2026-08-16_13-42-22
Die size	354.38 μm × 399.98 μm	354.01 μm × 399.61 μm
Std-cell area	41 678 μm^2	41 630 μm^2
Std-cell utilisation	35.5 %	36.2 %
Setup / hold WNS (typ 3.3 V 25 °C)	0 / 0	0 / 0
Detailed-routing DRC / antenna	0 / 0	0 / 0
Max slew / cap / fanout violations	0 / 0 / 0	0 / 0 / 0
LVS (Netgen)	match uniquely	match uniquely
Gate-level simulation	pass	pass, all 4 SPI modes
Signoff DRC (full IHP deck)	clean ¹	clean ¹

Gate-level simulation runs the RTL testbenches against the synthesised sg13g2_hv_* netlist using zero-delay functional models of the library cells — the check that proves the non-standard flip-flop mapping.

The first signoff DRC of these builds reported ~19 000 Cnt/CntB markers, which traced not to the flow but to a defect in the library itself: rail-tap contacts at cell-specific x positions, merging illegally wherever two *different* cells share a power rail — a geometry the library's own DRC harness never built. The fix (site-centred rail-tap grid) and the regression harness live in the library repository; both builds were re-run on the fixed library. The full analysis is the library's companion report, *The Shared-Rail Contact Clash in sg13g2_stdcell_hv*.

8.3 Comparing the two libraries

	Mode-0, 1.2 V	Mode-0, 3.3 V	All-modes, 1.2 V	All-modes, 3.3 V
Library	sg13g2_stdcell	sg13g2_stdcell_hv	sg13g2_stdcell	sg13g2_stdcell_hv

	Mode-0, 1.2 V	Mode-0, 3.3 V	All-modes, 1.2 V	All-modes, 3.3 V
Row height / tracks	3.78 μm / 9	7.14 μm / 17	3.78 μm / 9	7.14 μm / 17
Die size (μm)	157.78 $\times$ 176.50	354.38 $\times$ 399.98	160.61 $\times$ 179.33	354.01 $\times$ 399.61
Die area	27 848 μm^2	141 745 μm^2 (5.1$\times$)	28 802 μm^2	141 462 μm^2 (4.9$\times$)
Std-cell area	11 284 μm^2	41 678 μm^2 (3.7$\times$)	11 737 μm^2	41 630 μm^2 (3.5$\times$)
Utilisation	53.8 %	35.5 %	53.5 %	36.2 %
Clock closed	100 MHz	100 MHz	100 MHz	100 MHz
Timing corners	3 (ss/tt/ff)	1 (typ only)	3	1
Total power (tt)	$\approx$ 1.01 mW @ 1.2 V	not characterised	$\approx$ 1.08 mW @ 1.2 V	not characterised
Signoff DRC	clean ¹	clean ¹	clean ¹	clean ¹

The $\sim 3.6\times$ cell-area and $\sim 5\times$ die-area cost of the 3.3 V build follows directly from the library geometry — 1.89 $\times$ taller rows and 2.40 $\times$ wider PMOS — not from the design: cell count and netlist structure are essentially unchanged. Both libraries close the same 100 MHz clock with zero slack violations, so the SPI-side constraint is identical ($f_{SCK} \leq f_{CLK}/8 = 12.5$ MHz), and the lower utilisation of the thick-oxide builds is headroom, not congestion (routing closes with zero DRC at 35–36 %). What the 3.3 V build buys is native 3.3 V I/O — direct connection to the Raspberry Pi and AURIX hosts with no level shifting — at the price of single-corner timing sign-off and no power numbers, both library limitations rather than flow limitations.

9 Conclusions

- Both `spi_slave` variants are functionally verified at RTL and against their synthesised netlists; the all-modes variant is additionally verified in all four SPI modes against Raspberry Pi and AURIX QSPI master models, including a full register-map sweep at the maximum-rated SCK of $f_{CLK}/8$.
- Both variants complete the LibreLane v3 RTL-to-GDS flow **on both standard-cell libraries** — 1.2 V `sg13g2_stdcell` and 3.3 V `sg13g2_stdcell_hv` — with zero timing, antenna and routing violations, and pass Netgen LVS with uniquely matching circuits; the thick-oxide netlists additionally pass gate-level simulation, proving the non-standard flip-flop mapping.
- All four core macros pass the IHP SG13G2 signoff DRC deck; the only residual markers are the expected chip-level density/filler checks common to all unfilled

macros. The thick-oxide signoff surfaced, and drove the fix of, a shared-rail contact defect in the cell library (section 8.2).

- The thick-oxide build costs $\sim 3.6\times$ the cell area and $\sim 5\times$ the die area for the same 100 MHz closure, and buys native 3.3 V I/O toward both target hosts; its sign-off is weaker only where the library is (one timing corner, no power tables).
- The LibreLane v3 KLayout.SealRing step is defective (core overlap + wrong EdgeSeal marker) and its output is excluded from the deliverables. Seal-ring insertion, dummy fill, and final chip-level DRC remain tapeout-assembly tasks.